This work done by the force F during a small displacement dx stores as potential energy

$$U(x)=\int_0^x k\,x'\,dx'=\frac{1}{2}k\left(x'\right)^2\Big|_0^x=\frac{1}{2}kx^2 \quad (10.65)$$

From equation (10.22), we can substitute the value of force constant $k = m\,\omega^2$ in equation (10.65),

$$U(x)=\frac{1}{2}m\omega^2x^2 \quad (10.66)$$

where ω is the natural frequency of the oscillating system. For the particle executing simple harmonic motion from equation (10.6), we get

$$x = A \sin \omega t$$

$$U(\mathrm{t})=\frac{1}{2}m\omega^2A^2\sin^2\omega t \quad (10.67)$$

This variation of U is shown below.

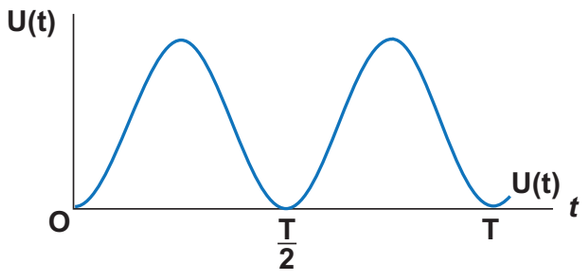


Figure 10.23 Variation of potential energy with time t

Question to think over

"If the potential energy is minimum then its second derivative is positive, why?"

b. Expression for Kinetic Energy

Kinetic energy

$$KE=\frac{1}{2}mv_x^2=\frac{1}{2}m\left(\frac{dx}{dt}\right)^2 \quad (10.68)$$

Since the particle is executing simple harmonic motion, from equation (10.6)

$$x = A \sin \omega t$$

Therefore, velocity is

$$v_x=\frac{dx}{dt}=A\omega\cos\omega t \quad (10.69)$$

$$=A\omega\sqrt{1-\left(\frac{x}{A}\right)^2}$$

$$v_x=\omega\sqrt{A^2-x^2} \quad (10.70)$$

Hence,

$$KE=\frac{1}{2}mv_x^2=\frac{1}{2}m\omega^2\left(A^2-x^2\right) \quad (10.71)$$

$$KE=\frac{1}{2}m\omega^2A^2\cos^2\omega t \quad (10.72)$$

This variation with time is shown below.

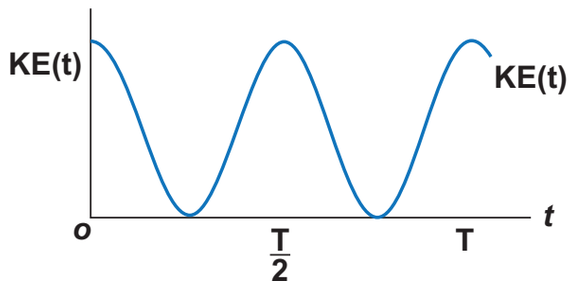


Figure 10.24 Variation of kinetic energy with time t.

c. Expression for Total Energy

Total energy is the sum of kinetic energy and potential energy

$$E = KE + U \quad (10.73)$$

$$E=\frac{1}{2}m\omega^2(\mathrm{A}^2-x^2)+\frac{1}{2}m\omega^2x^2$$

Hence, cancelling x^2 term,

$$E=\frac{1}{2}m\omega^2A^2=\text{ constant} \quad (10.74)$$

Alternatively, from equation (10.67) and equation (10.72), we get the total energy as

$$E=\frac{1}{2}m\omega^2A^2\sin^2\omega t+\frac{1}{2}m\omega^2A^2\cos^2\omega t$$

$$=\frac{1}{2}m\omega^2A^2(\sin^2\omega\mathrm{t}+\cos^2\omega t)$$

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